

Micro I, class 5

We continued our discussion of the *expenditure function*. It gives the minimum expenditure at prices p to be at least as well off as the indifference set indexed by the utility number, $\bar{u}$. The set of solutions to this (EM) problem at $(p, \bar{u})$ is denoted $h(p, \bar{u})$ and called either the *Hicksian* or *compensated* demand.

The (McKenzie-) Shepard Lemma asserts that, if the expenditure function is differentiable in p at a point p^0 , then $\nabla_p e(p^0, \bar{u}) = h(p^0, \bar{u})$. The fact is a consequence of the Milgrom-Segal (2002) envelope theorem. The Lemma implies that if we know something about the *second* derivative of $e(\cdot, \bar{u})$ then we learn something about the first derivative of $h(\cdot, \bar{u})$.

We proved that the expenditure function is, for each fixed $\bar{u}$, nondecreasing, homogenous of degree 1, and concave. If $e(\cdot, \bar{u})$, is C^2 , then these properties imply that the Hessian matrix of $e(\cdot, \bar{u})$ is a symmetric, negative semi-definite matrix. It follows by Shephard's Lemma that the Jacobian matrix of $h(\cdot, \bar{u})$ is symmetric, negative semi-definite matrix for every $p \gg 0$.

[The matrix $D_p h(p, \bar{u})$ is negative semidefinite if and only if

$$\sum_{i=1}^L \sum_{j=1}^L \frac{\partial h_i}{\partial p_j} v_i v_j \leq 0 \quad (1)$$

for every $v = (v_1, \dots, v_L) \in \mathbb{R}^L$.]

In particular, $\partial h_i(p, \bar{u})/\partial p_i \leq 0$ for $i = 1, \dots, L$ (a version of a law of demand) and $\partial h_i(p, \bar{u})/\partial p_j = \partial h_j(p, \bar{u})/\partial p_i$ for every i and j (symmetry of cross-price effects). [For the first, set $v_j = 0$ for $j \neq i$ in (1).]

At the end I introduced a Lemma asserting the equality between the Hicksian and ordinary demands, when evaluated at the appropriate arguments; to wit, $d(p, e(p, \bar{u})) = h(p, \bar{u})$. Next we will put all of this together to derive testable implications of the standard theory of the consumer. These are summarized in my note *Demand Properties: Summary*. I encourage you to read through these before Tuesday.